

The Girsanov Theorem

A Joint Report by

by

Arunav Bhowmick

Avinandan Roy

Indian Statistical Institute Kolkata

203 Barrackpore Trunk Rd, Kolkata, WB 700108, India.

1 Introduction

The Girsanov theorem is a very powerful and well-known result in the literature of stochastic calculus, particularly helpful in dealing with drifted Brownian motions. Let us first look at a motivating result. Consider $Y_1, Y_2, \dots, Y_n$ to be an i.i.d. sample from a non-increasing density f on $[0, \infty)$, and we want to estimate f . The log-likelihood of the observations is

$$\sum_{i=1}^n \log f(Y_i).$$

In order to estimate f , we try to maximize the log-likelihood over the class $\mathcal{G}$, given by

$$\mathcal{G} = \left\{ g : [0, \infty) \rightarrow [0, \infty), \int_0^\infty g(x) dx = 1, g \text{ is non-increasing} \right\}.$$

Then $\hat{f}_n = \arg \max_{g \in \mathcal{G}} n^{-1} \sum_{i=1}^n \log g(Y_i)$ is known as the Grenander estimator. The asymptotic distribution of the Grenander estimator was found by [Prakasa Rao \(1969\)](#). The following theorem describes the result.

Theorem 1.1 (Prakasa Rao, 1969). *Suppose that $f : [0, \infty) \rightarrow [0, \infty)$ has a strictly negative derivative f' at an interior point x_0 and $f(x_0) > 0$. Then,*

$$n^{1/3}(\hat{f}_n(x_0) - f(x_0)) \xrightarrow{d} |4f'(x_0)f(x_0)|^{1/3} \arg \max_{t \in \mathbb{R}} \{W_t - t^2\},$$

where $(W_t : t \in \mathbb{R})$ is a two-sided standard Brownian motion on $\mathbb{R}$.

Notice that the limiting distribution is a continuous functional of a drifted Brownian motion. We have seen results; for instance, in Donsker's theorem, where the limiting process is simply a

standard Brownian motion, which allows us to apply the continuous mapping theorem to find the distribution of the continuous functional. However, in this example, it is difficult to analyze the distribution of the drifted Brownian motion itself. In such cases, it becomes necessary to construct appropriate measures under which translated processes continue to be a standard Brownian motion. The Girsanov theorem enables us to construct such probability measures.

In order to motivate the construction of the new measure, we shall first look at a discrete analog. Consider independent standard normal random variables $Z_1, \dots, Z_n$ on $(\Omega, \mathcal{F}, P)$. Given a vector $(\mu_1, \dots, \mu_n) \in \mathbb{R}^n$, consider the new probability measure $\tilde{P}^{(\mu)}$ on $(\Omega, \mathcal{F})$ given by

$$\tilde{P}^{(\mu)}(d\omega) = \exp \left[\sum_{i=1}^n \mu_i Z_i(\omega) - \frac{1}{2} \sum_{i=1}^n \mu_i^2 \right] \cdot P(d\omega)$$

Then, $\tilde{P}^{(\mu)}[Z_1 \in dz_1, \dots, Z_n \in dz_n]$ is given by

$$\begin{aligned} & \exp \left[\sum_{i=1}^n \mu_i z_i - \frac{1}{2} \sum_{i=1}^n \mu_i^2 \right] \cdot P[Z_1 \in dz_1, \dots, Z_n \in dz_n] \\ &= \exp \left[\sum_{i=1}^n \mu_i z_i - \frac{1}{2} \sum_{i=1}^n \mu_i^2 \right] \times \frac{1}{(2\pi)^{n/2}} \exp \left[-\frac{1}{2} \sum_{i=1}^n z_i^2 \right] dz_1 \cdots dz_n \\ &= (2\pi)^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (z_i - \mu_i)^2 \right] dz_1 \cdots dz_n. \end{aligned}$$

Hence, under $\tilde{P}^{(\mu)}$, the random variables Z_i are independent normal random variables with $\tilde{E}^{(\mu)}[Z_i] = \mu_i$ and $\tilde{E}^{(\mu)}[(Z_i - \mu_i)^2] = 1$. So, $\{\tilde{Z}_i = Z_i - \mu_i : 1 \leq i \leq n\}$ are independent, standard normal random variables on $(\Omega, \mathcal{F}, \tilde{P}^{(\mu)})$. The Girsanov Theorem extends this idea to a continuous setting with d -dimensional Brownian motion.

2 Preliminaries

Before stating the main result, we introduce some necessary definitions and assumptions. We shall have our probability space $(\Omega, \mathcal{F}, P)$, where we consider the d -dimensional Brownian motion $W = \{W_t = (W_t^{(1)}, \dots, W_t^{(d)}), \mathcal{F}_t; 0 \leq t < \infty\}$, and the filtration $\{\mathcal{F}_t\}$ satisfies two conditions-

- a. it is right continuous,
- b. all the P -null sets of $\mathcal{F}$ are contained in $\mathcal{F}_0$.

Let $X = \{X_t = (X_t^{(1)}, \dots, X_t^{(d)}), \mathcal{F}_t; 0 \leq t < \infty\}$ be a vector of a measurable and adapted process which satisfies

$$E \left[\int_0^T (X_t^{(i)})^2 dt \right] < \infty; \quad 1 \leq i \leq d, \quad 0 \leq T < \infty. \quad (1)$$

The condition stated above is required to define the stochastic integrals of the adapted measurable processes $X^{(i)}$, with respect to the Brownian motions $W^{(i)}$. More generally, this condition is also needed to define the stochastic integrals of $X^{(i)}$, with respect to continuous square-integrable martingales. $\mathcal{M}_2^c$ denotes the space of all continuous square-integrable martingales. Define

$$[X]_T = E \int_0^T X_t^2 dt, \quad (2)$$

and let $\mathcal{L}$ be the space of all measurable processes X for which $[X]_T < \infty$ for all $T > 0$. Now, define $\mathcal{L}^*$ to be the space of all progressively measurable processes in $\mathcal{L}$, and on this space consider the metric

$$\|f\| = \left(E \int_0^\infty f(t)^2 dt \right)^{1/2}.$$

We have the approximation theorem there, by which we showed that $\mathcal{L}_0$, the space of all compactly supported measurable functions, is dense in $\mathcal{L}^*$. To make sense of stochastic integrals with respect to continuous square-integrable martingales, we need the **Doob-Meyer decomposition theorem**, which says that if X is a right-continuous submartingale adapted with filtration $\{\mathcal{F}_t\}$, we can uniquely decompose X as the sum of a right-continuous martingale $M = \{M_t, \mathcal{F}_t; 0 \leq t < \infty\}$ and an increasing process $A = \{A_t, \mathcal{F}_t; 0 \leq t < \infty\}$. Now, if X is a continuous square-integrable martingale ($\in \mathcal{M}_2^c$), then X^2 is a continuous submartingale. In that case, we can decompose it using Doob-Meyer, and the increasing part is defined as $\langle X \rangle = \{\langle X \rangle_t, \mathcal{F}_t; 0 \leq t < \infty\}$ and the martingale part is $M = \{X_t^2 - \langle X \rangle_t, \mathcal{F}_t; 0 \leq t < \infty\}$. Then, for $M \in \mathcal{M}_2^c$ and a measurable process Y , the stochastic integral is denoted in the following form

$$I_T^M(Y) = \int_0^T Y_t dM_t.$$

To make sense of these types of integrals, the quantity $[X]_T^2$ is defined as

$$[X]_T = E \int_0^T X_t^2 d\langle M \rangle_t, \quad (3)$$

and then in $\mathcal{L}^*$, the metric under which $\mathcal{L}_0$ becomes dense is defined as

$$[X] = \sum_{n=1}^{\infty} 2^{-n} (1 \wedge [X]_n). \quad (4)$$

For the approximation theorem for martingales in $\mathcal{M}_2^c$, we will need this following condition to hold:

$$[X]_T = E \int_0^T X_t^2 d\langle M \rangle_t < \infty, \quad (5)$$

while considering the stochastic integrals with respect to $M \in \mathcal{M}_2^c$. We need to define one more

quantity, which is $\|X\|$, for $X \in \mathcal{M}_2^c$. It is defined as

$$\|X\|_s^2 := E(X_s^2).$$

Here, we state some standard results without proof that will be used in the proof of the Girsanov theorem, which hold for $X, Y \in \mathcal{L}^*$.

$$I_0^M(X) = 0 \quad \text{a.s. } P \quad (6)$$

$$E[I_t^M(X) | \mathcal{F}_s] = I_s^M(X) \quad \text{a.s. } P \quad (7)$$

$$E[I_t^M(X)^2] = E \int_0^t X_u^2 d\langle M \rangle_u \quad (8)$$

$$\|I^M(X)\| = [X] \quad (9)$$

$$E \left[(I_t^M(X) - I_s^M(X))^2 | \mathcal{F}_s \right] = E \left[\int_s^t X_u^2 d\langle M \rangle_u \right] \quad (10)$$

$$I^M(\alpha X + \beta Y) = \alpha I^M(X) + \beta I^M(Y) \quad (11)$$

We shall need the cross-variation process. Cross-variation processes are defined as

$$\langle X, Y \rangle := \frac{1}{4} [\langle X + Y \rangle - \langle X - Y \rangle].$$

Then $XY - \langle X, Y \rangle$ becomes the martingale part of the process XY .

If condition (1) holds, $I^{W^{(i)}}(X^{(i)})$ is defined and becomes a member of $\mathcal{M}_2^c$, which follows from (6), (7) and (8) (since $E[I_t^{W^{(i)}}(X)^2] = E \int_0^t X_u^2 d\langle W^{(i)} \rangle_u = E \int_0^t X_u^2 du < \infty$). Now, we shall proceed towards constructing the new probability measure. For that let us define a process $Z(X)$ as

$$Z_t(X) = \exp \left[\sum_{i=1}^d \int_0^t X_s^{(i)} dW_s^{(i)} - \frac{1}{2} \int_0^t \|X\|_s^2 ds \right], \quad 0 \leq t < \infty. \quad (12)$$

For the time being, we will assume that $Z(X) \in \mathcal{M}_2^c$ (we shall later state a sufficient condition under which the process $Z(X)$ will be a martingale). Then, $E[Z_t(X)] = 1$ for all $t \in [0, \infty)$ since $Z_0(X) = 1$. In that case, we define for each $0 \leq T < \infty$, the new probability measure $\tilde{P}_T$ on $(\Omega, \mathcal{F}_T)$ as

$$\tilde{P}_T(A) := E[1_A Z_T(X)], \quad A \in \mathcal{F}_T. \quad (13)$$

We present a result describing the consistency of the family $\{\tilde{P}_T; 0 \leq T < \infty\}$.

Proposition 2.1. *The family of probability measures $\{\tilde{P}_T; 0 \leq T < \infty\}$ satisfies the following property*

$$\tilde{P}_T(A) = \tilde{P}_t(A); \quad A \in \mathcal{F}_t, \quad 0 \leq t \leq T. \quad (14)$$

The proof of the above proposition follows directly from the martingale property of $Z(X)$. We will need some additional results before we state and prove the main theorem.

Lemma 2.2. *Let $\tilde{E}_T$ be the expectation operator with respect to the probability measure $\tilde{P}_T$. Assume $Z(X)$ to be a martingale. If $0 \leq s \leq t \leq T$ and Y is an $\mathcal{F}_t$ measurable function such that $\tilde{E}_T[|Y|] < \infty$, then we have the following Bayes' rule:*

$$\tilde{E}_T[Y|\mathcal{F}_s] = \frac{1}{Z_s(X)} E[Y Z_t(X)|\mathcal{F}_s] \quad \text{a.s. } P \text{ and } \tilde{P}_T. \quad (15)$$

Proof. Consider $A \in \mathcal{F}_s$. Then,

$$\begin{aligned} & \tilde{E}_T \left[\mathbf{1}_A \frac{1}{Z_s(X)} E[Y Z_t(X)|\mathcal{F}_s] \right] \\ &= \tilde{E}_s \left[\mathbf{1}_A \frac{1}{Z_s(X)} E[Y Z_t(X)|\mathcal{F}_s] \right] && \text{(using proposition 2.1)} \\ &= E \left[\mathbf{1}_A E[Y Z_t(X)|\mathcal{F}_s] \right] && \text{(using definition of } \tilde{P}_s) \\ &= E \left[\mathbf{1}_A Y Z_t(X) \right] && \text{(using definition of conditional expectations)} \\ &= \tilde{E}_t[\mathbf{1}_A Y] && \text{(using definition of } \tilde{P}_t) \\ &= \tilde{E}_T[\mathbf{1}_A Y]. && \text{(using proposition 2.1)} \end{aligned}$$

Therefore, we have proved the a.s. equality in $\tilde{P}_T$ and P as required. $\square$

Finally, we state and prove a martingale characterization theorem for Brownian motions which will crucially be used in the proof of the Girsanov theorem.

Theorem 2.3 (Levy, 1948). *Let $X = \{X_t = (X_t^{(1)}, \dots, X_t^{(d)}), \mathcal{F}_t; 0 \leq t < \infty\}$ be a continuous, adapted process in $\mathbb{R}^d$ such that for all $1 \leq i \leq d$, $X_0^{(i)} = 0$ a.s. and $X^{(i)}$ is a martingale relative to $\{\mathcal{F}_t\}$. If the cross-variations are given by*

$$\langle X^{(i)}, X^{(j)} \rangle_t = \delta_{ij}t, \quad 1 \leq i, j \leq d, \quad (16)$$

then $\{X_t, \mathcal{F}_t; 0 \leq t < \infty\}$ is a d -dimensional Brownian motion.

Remark. The original theorem is stated for continuous local martingales. This is a weaker version of the original result.

Proof. To show that $\{X_t, \mathcal{F}_t; 0 \leq t < \infty\}$ is a d -dimensional Brownian motion, it suffices to show that for $0 \leq s < t$, $X_t - X_s$ is independent of $\mathcal{F}_s$ and distributed as a d -variate normal random vector with mean zero and covariance matrix $(t - s)I_d$. By uniqueness of characteristic functions,

it is enough to show that for any $u \in \mathbb{R}$,

$$E \left[\exp(iu^\top (X_t - X_s)) \mid \mathcal{F}_s \right] = \exp \left(-\frac{\|u\|^2}{2} (t - s) \right), \quad \text{a.s. } P. \quad (17)$$

By the multi-dimensional version of Ito's rule ([Karatzas and Shreve, 2012](#)), we have for any $f \in C^2(\mathbb{R}^d)$,

$$f(X_t) = f(0) + \sum_{j=1}^d \int_0^t \frac{\partial}{\partial x_j} f(X_v) dX_v^{(j)} + \frac{1}{2} \sum_{j=1}^d \sum_{k=1}^d \int_0^t \frac{\partial^2}{\partial x_j \partial x_k} f(X_v) d\langle M^{(j)}, M^{(k)} \rangle_v, \quad 0 \leq t < \infty,$$

where we have utilized the fact that $X_0 = 0$ a.s. and $X^{(j)}$ is a martingale for each $1 \leq j \leq d$. Applying this formula for the function $f(x) = \exp(iu^\top x)$, we get

$$\exp(iu^\top X_t) = 1 + i \sum_{j=1}^d u_j \int_0^t \exp(iu^\top X_v) dX_v^{(j)} - \frac{1}{2} \|u\|^2 \int_0^t \exp(iu^\top X_v) dv,$$

where (16) is used to simplify the last term on the right hand side. We get a similar expression by replacing t with s , and by subtracting the latter from the former, we obtain

$$\exp(iu^\top X_t) - \exp(iu^\top X_s) = i \sum_{j=1}^d u_j \int_s^t \exp(iu^\top X_v) dX_v^{(j)} - \frac{1}{2} \|u\|^2 \int_s^t \exp(iu^\top X_v) dv. \quad (18)$$

Since $\langle X^{(j)} \rangle_t = t$, we have $X^{(j)} \in \mathcal{M}_2^c$. Combining this with the fact that $|\exp(iu^\top x)| \leq 1$, we conclude that

$$\left\{ \int_0^t \exp(iu^\top X_v) dX_v^{(j)}, \mathcal{F}_t; 0 \leq t < \infty \right\} \in \mathcal{M}_2^c,$$

for each $1 \leq j \leq d$. As a result of this,

$$E \left[\int_s^t \exp(iu^\top X_v) dX_v^{(j)} \mid \mathcal{F}_s \right] = 0, \quad \text{a.s. } P.$$

Multiplying by $\exp(-iu^\top X_s) I_A$ throughout in (18), where $A \in \mathcal{F}_s$, and taking expectation afterwards, we get

$$E[\exp(iu^\top (X_t - X_s)) \mathbf{1}_A] - P(A) = -\frac{1}{2} \|u\|^2 \int_s^t E[\exp(iu^\top (X_v - X_s)) \mathbf{1}_A] dv.$$

Define the function $g(v) = E[\exp(iu^\top (X_v - X_s)) \mathbf{1}_A]$, and observe that it satisfies the integral equation

$$g(t) = g(s) - \frac{1}{2} \|u\|^2 \int_s^t g(v) dv, \quad t \geq s \quad \text{and} \quad g(s) = P(A).$$

The solution to this integral equation is given by

$$g(v) = P(A) \exp \left(-\frac{\|u\|^2}{2}(v-s) \right),$$

and this implies that

$$E \left[\exp (iu^\top (X_t - X_s)) \mathbf{1}_A \right] = E \left[\exp \left(-\frac{\|u\|^2}{2}(t-s) \right) \mathbf{1}_A \right], \quad \text{for all } A \in \mathcal{F}_s,$$

and (17) follows. $\square$

3 Girsanov Theorem

Theorem 3.1 (Girsanov, 1960). *Assume that $Z(X)$ defined by (12) is a martingale. Define a process $\widetilde{W} = \{\widetilde{W}_t = (\widetilde{W}_t^{(1)}, \dots, \widetilde{W}_t^{(d)}), \mathcal{F}_t; 0 \leq t < \infty\}$ by*

$$\widetilde{W}_t^{(i)} := W_t^{(i)} - \int_0^t X_s^{(i)} ds, \quad 1 \leq i \leq d, \quad 0 \leq t < \infty. \quad (19)$$

For each fixed $T \in [0, \infty)$, the process $\{\widetilde{W}_t, \mathcal{F}_t; 0 \leq t \leq T\}$ is a d -dimensional Brownian motion on $(\Omega, \mathcal{F}_T, \widetilde{P}_T)$.

Proof. Fix $0 \leq T < \infty$ for the entirety of the proof. We start by claiming that $\widetilde{W}^{(j)} \in \widetilde{\mathcal{M}}_{2,T}^c$, the space of all continuous square-integrable martingales in $(\Omega, \mathcal{F}_T, \widetilde{P}_T)$, for all $j = 1, \dots, d$. To show this, first note that $W^{(j)}$ is a Brownian motion on $(\Omega, \mathcal{F}, P)$ for all $j = 1, \dots, d$. Fix $1 \leq j \leq d$ and $0 \leq s \leq T$. Note that $W_t^{(j)}$ is $\mathcal{F}_t$ measurable, $W^{(j)} \in \mathcal{M}_2^c$, and

$$\widetilde{E}_T \left[|W_t^{(j)}| \right] = E \left[|W_t^{(j)}| Z_T(X) \right] \leq \left(E \left[W_t^{(j)} \right]^2 \right)^{\frac{1}{2}} \left(E \left[Z_T(X) \right]^2 \right)^{\frac{1}{2}} < \infty.$$

Then by Lemma 2.2, we can say that

$$\widetilde{E}_T \left[W_t^{(j)} | \mathcal{F}_s \right] = \frac{1}{Z_s(X)} E \left[W_t^{(j)} Z_t(X) | \mathcal{F}_s \right] = W_s^{(j)} \quad a.s. \quad P \text{ and } \widetilde{P}_T. \quad (20)$$

Therefore, $W^j \in \widetilde{\mathcal{M}}_{2,T}^c$. Denote $N_t = \sum_{i=1}^d \int_0^t X_s^{(i)} dW_s^{(i)}$ and $B_t = -\frac{1}{2} \int_0^t \|X\|_s^2 ds$. Then, using the conventions $(\Delta t)^2 = (\Delta t)(\Delta W_t) = 0$ and $(\Delta W_t)^2 = \Delta t$, we have

$$\begin{aligned} (i) \quad \Delta N_t &= \sum_{i=1}^d X_t^{(i)} \Delta W_t^{(i)} \implies (\Delta N_t)^2 = -\Delta B_t \\ (ii) \quad \Delta B_t &= -\frac{1}{2} \|X\|_t^2 \Delta t \\ \therefore \quad Z_{t+\Delta t}(X) - Z_t(X) &= Z_t(X) (\Delta N_t + \Delta B_t + \frac{1}{2} (\Delta N_t)^2) = Z_t(X) \Delta N_t \end{aligned}$$

$$\implies Z_{t+\Delta t} = Z_t(X)(1 + \Delta N_t).$$

From here, we obtain

$$Z_t(X) = Z_0(X) + \sum_{i=1}^d \int_0^t Z_s(X) X_s^{(i)} dW_s^{(i)}. \quad (21)$$

Notice that

$$\widetilde{W}_{t+\Delta t}^{(j)} = \widetilde{W}_t^{(j)} + \Delta W_t^{(j)} - X_t^{(j)} \Delta t.$$

Hence, we get

$$\begin{aligned} & Z_{t+\Delta t}(X) \widetilde{W}_{t+\Delta t}^{(j)} - Z_t(X) \widetilde{W}_t^{(j)} \\ &= Z_t(X)(1 + \Delta N_t)(\widetilde{W}_t^{(j)} + \Delta W_t^{(j)} - X_t^{(j)} \Delta t) - Z_t(X) \widetilde{W}_t^{(j)} \\ &= Z_t(X) \Delta W_t^{(j)} - Z_t(X) X_t^{(j)} \Delta t + Z_t(X) \Delta N_t \Delta \widetilde{W}_t^{(j)} + Z_t(X) \widetilde{W}_t^{(j)} \Delta N_t \\ &= Z_t(X) \Delta W_t^{(j)} + \sum_{i=1}^d Z_t(X) \widetilde{W}_t^{(j)} X_t^{(i)} \Delta W_t^{(i)}. \end{aligned}$$

So, we get the representation

$$Z_t(X) \widetilde{W}_t^{(j)} = \int_0^t Z_s(X) dW_s^{(j)} + \sum_{i=1}^d \int_0^t Z_s(X) \widetilde{W}_s^{(j)} X_s^{(i)} dW_s^{(i)}. \quad (22)$$

Therefore, $Z_t(X) \widetilde{W}_t^{(j)}$ is a martingale in $(\Omega, \mathcal{F}, P)$. To prove the boundedness of $Z_t(X) \widetilde{W}_t^{(j)}$, we note that

$$\begin{aligned} & E \left[|Z_t(X) \widetilde{W}_t^{(j)}| \right] \\ &= E \left[\left| Z_t(X) W_t^{(j)} - Z_t(X) \int_0^t X_s ds \right| \right] \\ &\leq E \left[|Z_t(X) W_t^{(j)}| \right] + E \left[\left| Z_t(X) \int_0^t X_s ds \right| \right] \\ &\leq \|Z_t(X)\| \cdot \left(\|W_t^{(j)}\| + E \left[\left| \int_0^t X_s ds \right|^2 \right]^{1/2} \right) \quad (\text{using C.S. inequality}) \\ &\leq \|Z_t(X)\| \cdot \left(\sqrt{t} + \sqrt{t} E \left[\int_0^t X_s^2 ds \right]^{1/2} \right) \quad (\text{using C.S. inequality}) \\ &< \infty. \end{aligned}$$

Consequently, we also get that $\tilde{E}_T[\|\widetilde{W}_t^{(j)}\|] < \infty$. Hence, we can apply Lemma 2.2 to conclude that

$$\tilde{E}_T \left[\widetilde{W}_t^{(j)} | \mathcal{F}_s \right] = \frac{1}{Z_s(X)} E[Z_t(X) \widetilde{W}_t^{(j)}] = \widetilde{W}_s^{(j)} \quad a.s. \quad P \text{ and } \tilde{P}_T.$$

Thus, $\widetilde{W}^{(j)} \in \mathcal{M}_{2,T}^c$ and the claim is proven.

Next, we compute the cross-variations. We can calculate the Ito rule for $\widetilde{W}_t^{(j)}\widetilde{W}_t^{(k)}$ and the calculation is as follows

$$\begin{aligned} & \widetilde{W}_{t+\Delta t}^{(j)}\widetilde{W}_{t+\Delta t}^{(k)} - \widetilde{W}_t^{(j)}\widetilde{W}_t^{(k)} \\ &= (\widetilde{W}_t^{(j)} + \Delta W_t^{(j)} - X_t^{(j)}\Delta t)(\widetilde{W}_t^{(k)} + \Delta W_t^{(k)} - X_t^{(k)}\Delta t) - \widetilde{W}_t^{(j)}\widetilde{W}_t^{(k)} \\ &= \widetilde{W}_t^{(j)}\Delta W_t^{(k)} - \widetilde{W}_t^{(j)}X_t^{(k)}\Delta t + \widetilde{W}_t^{(k)}\Delta W_t^{(j)} - X_t^{(j)}\widetilde{W}_t^{(k)}\Delta t + \delta_{jk}\Delta t. \end{aligned}$$

Hence, we get that

$$\begin{aligned} \widetilde{W}_t^{(j)}\widetilde{W}_t^{(k)} - \langle W^{(j)}, W^{(k)} \rangle_t &= \int_0^t \widetilde{W}_s^{(j)}dW_s^{(k)} + \int_0^t \widetilde{W}_s^{(k)}dW_s^{(j)} - \left[\int_0^t \widetilde{W}_s^{(j)}X_s^{(k)}ds \right. \\ &\quad \left. + \int_0^t \widetilde{W}_s^{(k)}X_s^{(j)}ds \right]. \end{aligned} \quad (23)$$

Note that while calculating $Z_t(X)\widetilde{W}_t^{(j)}$, we notice that (22) involves only stochastic integrals. This means that we can perform a similar calculation for the process $\widetilde{W}^{(j)}\widetilde{W}^{(k)} - \langle W^{(j)}, W^{(k)} \rangle$,

$$\begin{aligned} Z_t(X)[\widetilde{W}_t^{(j)}\widetilde{W}_t^{(k)} - \langle W^{(j)}, W^{(k)} \rangle_t] &= \int_0^t Z_s(X)\widetilde{W}_s^{(j)}dW_s^{(k)} + \int_0^t Z_s(X)\widetilde{W}_s^{(k)}dW_s^{(j)} \\ &\quad + \sum_{i=1}^d \int_0^t (\widetilde{W}_s^{(j)}\widetilde{W}_s^{(k)} - \langle W^{(j)}, W^{(k)} \rangle_s)Z_s(X)X_s^{(i)}dW_s^{(i)}. \end{aligned} \quad (24)$$

We can thus conclude that the process $Z(X)[\widetilde{W}^j\widetilde{W}^k - \langle W^{(j)}, W^{(k)} \rangle]$ is a martingale in $(\Omega, \mathcal{F}, P)$. Applying Lemma 2.2 again, we get for $0 \leq s \leq t \leq T$,

$$\widetilde{E}_T \left[\widetilde{W}_t^{(j)}\widetilde{W}_t^{(k)} - \langle W^{(j)}, W^{(k)} \rangle_t \middle| \mathcal{F}_s \right] = \widetilde{W}_s^{(j)}\widetilde{W}_s^{(k)} - \langle W^{(j)}, W^{(k)} \rangle_s \quad a.s. \quad P \text{ and } \widetilde{P}_T.$$

From uniqueness of Doob-Meyer decomposition, we obtain

$$\langle \widetilde{W}^{(j)}, \widetilde{W}^{(k)} \rangle_t = \langle W^{(j)}, W^{(k)} \rangle_t = \delta_{jk}t; \quad 0 \leq t \leq T.$$

Since all the criteria of Theorem 2.3 is satisfied, $\{\widetilde{W}_t, \mathcal{F}_t; 0 \leq t \leq T\}$ is a d -dimensional Brownian motion on $(\Omega, \mathcal{F}_T, \widetilde{P}_T)$. $\square$

In some applications, one wants to define $\widetilde{W}$ as a process on entire $[0, \infty)$ and for this purpose, the measures $\{\widetilde{P}_T : 0 \leq T < \infty\}$ are inadequate. What one wants is a single measure $\widetilde{P}$ defined on $\mathcal{F}_\infty$ which agrees with $\widetilde{P}_T$ when restricted to $\mathcal{F}_T$, but this may not exist in general. To overcome this difficulty, we restrict our attention to the probability space $(C[0, \infty)^d, \mathcal{B}(C[0, \infty)^d), P_W)$, where P_W is the Wiener measure. Let W be the co-ordinate mapping process, which is a d -dimensional

Brownian motion under P_W . Then we have the following result.

Corollary 3.2. *Let $X = \{X_t, \mathcal{F}_t^W; 0 \leq t < \infty\}$ be a d -dimensional progressively measurable satisfying (1). If $Z(X)$ of (12) is a martingale, then there is a unique probability measure $\tilde{P}$ satisfying*

$$\tilde{P}(A) = E[\mathbf{1}_A Z_t(X)], \quad A \in \mathcal{F}_t^W, \quad 0 \leq t < \infty,$$

and $\tilde{W}$ defined by (19) is a d -dimensional Brownian motion on $(C^d[0, \infty), \mathcal{F}_\infty^W, \tilde{P})$.

Throughout the presentation, we have assumed that $Z(X)$ is a martingale. We provide a sufficient condition to verify when this is indeed true.

Theorem 3.3 (Novikov, 1972). *Let $W = \{W_t = (W_t^{(1)}, \dots, W_t^{(d)}), \mathcal{F}_t; 0 \leq t < \infty\}$ be a d -dimensional Brownian motion and $X = \{X_t = (X_t^{(1)}, \dots, X_t^{(d)}), \mathcal{F}_t; 0 \leq t < \infty\}$ be a vector of measurable, adapted processes satisfying (1). If*

$$E \left[\exp \left(\frac{1}{2} \int_0^T \|X_s\|^2 ds \right) \right] < \infty, \quad 0 \leq T < \infty,$$

then $Z(X)$ is a martingale.

4 An Application

Consider a Brownian motion $W = \{W_t, \mathcal{F}_t; 0 \leq t < \infty\}$ and define

$$T_{a,b} = \inf\{t > 0 : W_t = a + bt\}$$

to be the first time W hits the line $a + bt$. We want to find out the probability that the line is hit by W , that is, we want to compute $P(T_{a,b} < \infty)$.

One possible approach to tackle this problem is to consider the first passage time of the drifted process $(W_t - bt : 0 \leq t < \infty)$ to the level a . So, we wish to construct a measure under which the drifted process $(W_t - bt : 0 \leq t < \infty)$ can be analyzed easily. Observe that by choosing X such that $X_t = -b$ for all $0 \leq t < \infty$, we get

$$Z_t = \exp(-bW_t - b^2t/2), \quad 0 \leq t < \infty.$$

By Corollary 3.2, we may assume without loss of generality that the process $\tilde{W} = \{\tilde{W}_t = W_t + bt, \mathcal{F}_t^W; 0 \leq t < \infty\}$ is a Brownian motion under the unique measure $P^{(b)}$ which satisfies

$$P^{(b)}(A) = E[\mathbf{1}_A Z_t], \quad A \in \mathcal{F}_t^W.$$

The distribution of $(W_t = \tilde{W}_t - bt : 0 \leq t < \infty)$ under $P^{(b)}$ is the same as the distribution of

$(W_t - bt : 0 \leq t < \infty)$ under P . Thus, we have

$$P(T_{a,b} \leq t) = P^{(b)}(T_a \leq t), \quad \text{for all } t \in \mathbb{R},$$

where $T_a = \inf\{t > 0 : W_t = a\}$. Note that since

$$\{T_a \leq t\} \in \mathcal{F}_t^W \cap \mathcal{F}_{T_a}^W = \mathcal{F}_{t \wedge T_a}^W,$$

we can write for fixed $t \in \mathbb{R}$,

$$\begin{aligned} P^{(b)}(T_a \leq t) &= E[\mathbf{1}_{\{T_a \leq t\}} Z_t] \\ &= E[E[\mathbf{1}_{\{T_a \leq t\}} Z_t \mid \mathcal{F}_{t \wedge T_a}^W]] = E[\mathbf{1}_{\{T_a \leq t\}} E[Z_t \mid \mathcal{F}_{t \wedge T_a}^W]] \\ &= E[\mathbf{1}_{\{T_a \leq t\}} Z_{t \wedge T_a}] = E[\mathbf{1}_{\{T_a \leq t\}} Z_{T_a}] \\ &= E[\mathbf{1}_{\{T_a \leq t\}} \exp(-ba - b^2 T_a / 2)]. \end{aligned}$$

Letting $t \rightarrow \infty$ in the last equality, we obtain

$$P^{(b)}(T_a < \infty) = \exp(-ba) E[-b^2 T_a / 2] = \exp(-ba - |ba|).$$

For $ab \leq 0$, $P(T_{a,b} \leq t) = 1$ and for $ab > 0$, $P(T_{a,b} \leq t) = \exp(-2ab) < 1$, i.e., $T_{a,b}$ is a defective random variable and the line is not hit with non-zero probability.

References

- Karatzas, I. and Shreve, S. (2012). *Brownian motion and stochastic calculus*, volume 113. Springer Science & Business Media.
- Prakasa Rao, B. L. S. (1969). Estimation of a unimodal density. *Sankhyā: The Indian Journal of Statistics, Series A*, 31:23–36.